

6.1 Waveforms, Spectrograms, and Time–Frequency Tokens

STFT frames, not 16 kHz samples. Patch the picture the way ViT patched pixels.

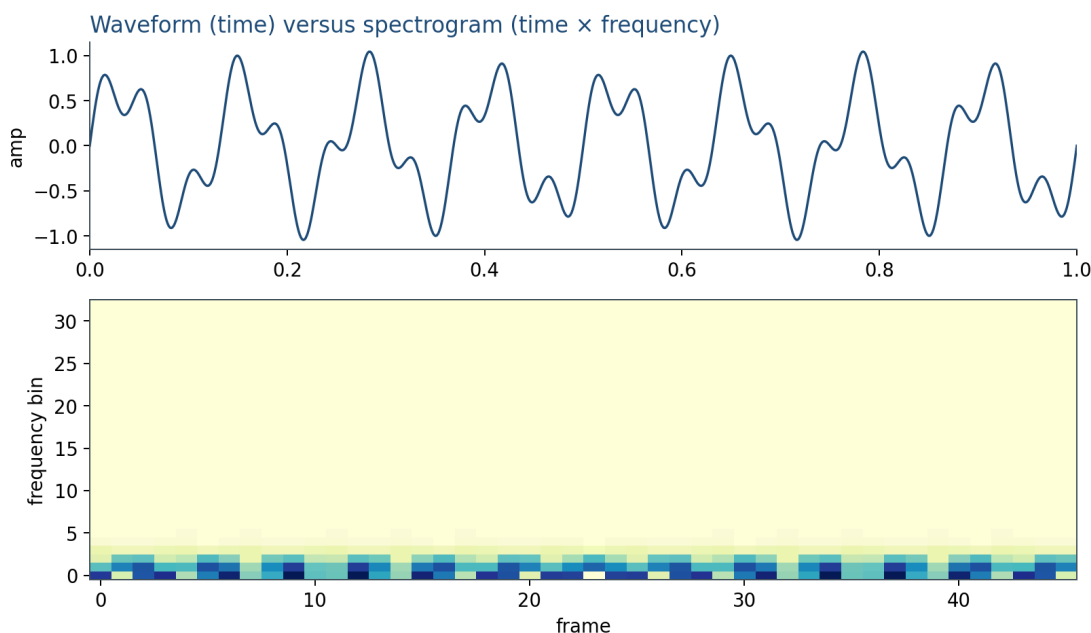
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A waveform is a 1-D list of samples. A **spectrogram** is that list cut into short windows, each turned into a frequency snapshot. Once you have a 2-D time–frequency picture, every vision trick from Week 4 (patches, a transformer, CLIP-style towers) applies to audio. Lab 6 will cut a sine mixture by hand; this hour is why the picture exists.

1. What a spectrogram is

A **spectrogram** is a picture of sound. Models do not eat 16 kHz samples as a raw list. You record pressure $x[n]$ at a sampling rate f_s (16 kHz is a common speech default). A sine of f hertz is $x[n] = \sin(2\pi f n / f_s)$. Mixtures add; that is still one stream.

The **short-time Fourier transform (STFT)** windows the signal (Hann, hop H , FFT size N), then takes a DFT of each frame. The **magnitude spectrogram** $S(t, f) = |X(t, f)|$ is what you plot. Phase exists, but most audio encoders drop it and keep magnitudes or a **mel** compression of them (log-mel filterbank). A **mel** spectrogram warps frequency to a perceptual scale.



A tall band that stays put is a steady tone. A sweep that climbs is a chirp. Speech looks like stacked formants that move.

Log-mel is a compressive front end: fewer bands than a linear STFT, spaced the way hearing is. It is still a spectrogram. Do not skip the plot; if the 440 Hz band is missing, the rest of the pipeline is theater.

2. Why we use it

Waveforms are long and oscillatory. A transformer on 16 kHz samples would see 16,000 tokens per second. An STFT with a 25 ms window and a 10 ms hop sees 100 frames per second. Those frames last long enough to see a phoneme, not a single pressure sample. A 2-D time–frequency map can then be patched like a ViT image.

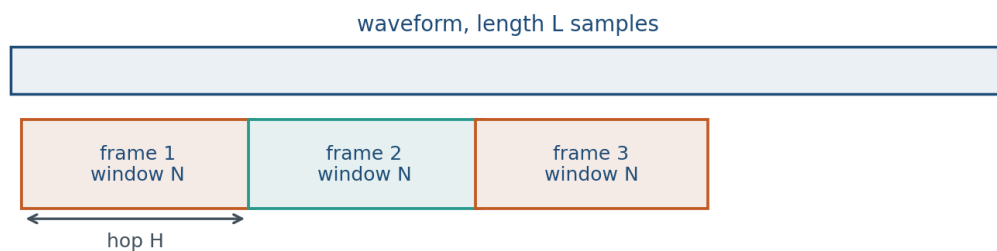
You cannot make time resolution and frequency resolution arbitrarily fine at once. Hop H is the time step; N is the frequency resolution. Small H means more frames (and more tokens later); large N means narrower Hertz bins and a longer window that smears onsets.

If your project uses audio, write f_s , window, hop, and whether you used mel or raw STFT. Those choices change the tokens.

3. Architecture

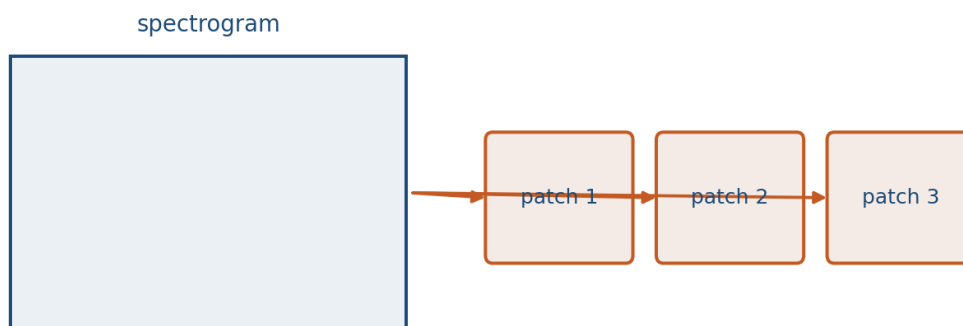
The boxes connect in that order: waveform, then frames, then STFT magnitudes (optionally mel + log), then patch embed, then transformer.

STFT: slide a window of length N by hop H . Frames $T = 1 + \lfloor (L - N)/H \rfloor$.



Treat S as a single-channel image. A CNN can ingest it. A ViT can **patch** it: tiles of time $\times$ frequency, flatten, linear map to width d , add positions. Lab 6 does that cut by hand.

Same trick as ViT: cut time-frequency patches, then embed



Time is not quite space. Adjacent frequency bins are related (harmonics); adjacent frames are related (smooth speech). Positions should know which axis is which. Still, "patch then attend" is the default modern front end.

Non-overlapping $p \times p$ tiles on an $F \times T$ grid give $(F / p) \times (T / p)$ tokens if both axes divide. Remainders are padded or cropped; Lab 6 crops.

This front end is not a transcriber. Whisper (note **6.2**) will decode text from this picture. CLAP will embed the picture next to a caption.

4. How it works, step by step

Lab 6: sine mixture, STFT, patch. Hop length and window are method knobs.

1. **Record.** Length L samples at rate f_s . Duration in seconds is L / f_s , not the frame count.
2. **Window.** Place frames of length N with hop H . Hann is the usual taper.
3. **FFT each frame.** Keep magnitudes, or a log-mel filterbank. Phase is usually dropped.
4. **Check a known tone.** Bin k of an N -point FFT sits near $k \cdot f_s / N$ hertz (real FFT: $k = 0, \dots, N / 2$). That conversion is how you check Lab 6's two sines.
5. **Patch.** Cut the $F \times T$ grid into non-overlapping tiles, flatten, linear map, add positions that know which axis is frequency.
6. **Hand the tokens to a transformer.** Encoding, retrieval, or transcription is a later head, not this picture.

A larger FFT does not "hear more of the clip." It hears a **longer window**, so onsets blur; hop still sets the time grid.

5. Mathematical formulas

Frame m of the STFT:

$$X(m, \omega) = \sum_n x[n]w[n - mH]e^{-j\omega n}.$$

The plot is usually $S(m, \omega) = |X(m, \omega)|$, or a mel filterbank of those magnitudes. Frame count, without padding, is

$$T = 1 + \left\lfloor \frac{L - N}{H} \right\rfloor$$

when the length $L \geq N$. Sequence length for the transformer is then about T frames, or the number of patches of those frames.

Bin k sits near $k \cdot f_s / N$ hertz. Non-overlapping $p \times p$ tiles on an $F \times T$ grid give $(F / p) \times (T / p)$ tokens when both axes divide.

6. Positive points and negative points

Positive.

- A spectrogram shares ViT code: patch, position, attend.
- 100 frames per second is a phoneme-scale sequence; 16,000 samples per second is not.
- Lab 6 can plot two sines and check the Hertz bins by hand.

Negative.

- Phase is often dropped; reconstruction is not the job of this encoder.
- Time resolution versus frequency resolution is a real tradeoff, not a software default.
- A spectrogram is not yet a transcriber, a speaker id, or a music tag (next note).
- Patching $F \times T$ as if it were a square image forgets which axis is frequency; remainders can drop a formant band if you crop carelessly.

When not to. Do not feed raw 16 kHz samples to a transformer and call the spectrogram optional. Do not count samples as frames.

7. What this is not

A spectrogram is not a transcript, a speaker id, or a music tag. Those are **tasks** on top of the tokens. Whisper (note **6.2**) will decode text from this picture. CLAP will embed the picture next to a caption. If your project uses audio, write f_s , window, hop, and whether you used mel or raw STFT. Those choices change the tokens.

Bin k of an N -point FFT sits near $k \cdot f_s / N$ hertz (real FFT: $k = 0, \dots, N / 2$). That conversion is how you check Lab 6's two sines.

8. Teaching this note

About **35 minutes** at the board in the two-hour week: draw one second of a 440 Hz sine, mark a window of length N and hop H , count frames, then cut a toy mel grid into patches. Play the STFT video (~10–12 min; see Video). Lab 6 (sine mixture, STFT, concat vs add) is studio time after notes **6.2–6.3**, not this block.

9. Worked example

One second of speech at $f_s = 16$ kHz: $L = 16000$ samples. Take $N = 400$, hop $H = 160$. Window $400 / 16000 = 25$ ms, hop 10 ms. Those are **Whisper's default** front-end numbers (note **6.2**).

$$T = 1 + \left\lfloor \frac{16000 - 400}{160} \right\rfloor = 1 + 97 = 98$$

frames. Each real FFT has $N / 2 + 1 = 201$ bins. A 440 Hz tone sits near bin $k \approx 440 \cdot 400 / 16000 = 11$.

Double the hop to $H = 320$ and keep N fixed:

$$T' = 1 + \left\lfloor \frac{15600}{320} \right\rfloor = 1 + 48 = 49.$$

About half as many frames; each frame is twice as far apart in time (20 ms). Frequency bins did not change, because N did not.

Now patch a log-mel grid of shape 80×400 with non-overlapping 16×16 tiles (ignore remainder):

$$\frac{80}{16} \times \frac{400}{16} = 5 \times 25 = 125$$

tokens. That is the number Lab 6 is practicing in miniature, and the number a Whisper-style encoder would attend over.

10. Where students get stuck

- Counting **samples** as **frames**. L / f_s is duration in seconds; T is how many windows you placed.
 - Thinking a larger FFT “hears more of the clip.” It hears a **longer window**, so onsets blur; hop still sets the time grid.
 - Patching $F \times T$ as if it were a square image and forgetting which axis is frequency. Remainders: crop in the lab; do not silently drop a formant band.
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11. Video

[Valerio Velardo — Short-Time Fourier Transform Explained Easily](#). Play the STFT / spectrogram walkthrough in class (~10–12 min). Pause when the window slides: that hop is H in the formula above.

Search “**Valerio Velardo Short-Time Fourier Transform**” if you need the rest of the Sound of AI series. The board hour is the frame count and the patch count, not the complex-phase algebra.

12. Practice

1. You double the hop length and keep N fixed. What happens to the number of time frames, and to time resolution?
2. Why is a spectrogram a better input to a transformer than the raw waveform at 16 kHz?
3. Count patch tokens for a mel grid of 80×400 with 16×16 tiles (ignore remainder).
4. $f_s = 8000$, one second, $N = 256$, hop $H = 128$. How many STFT frames? What Hertz spacing is one FFT bin (f_s / N)?
5. Same clip as (4), but you pad so the last frame is included and $T = \lceil L / H \rceil$. What is T , and why might a library report that instead of $1 + \lfloor (L - N) / H \rfloor$?